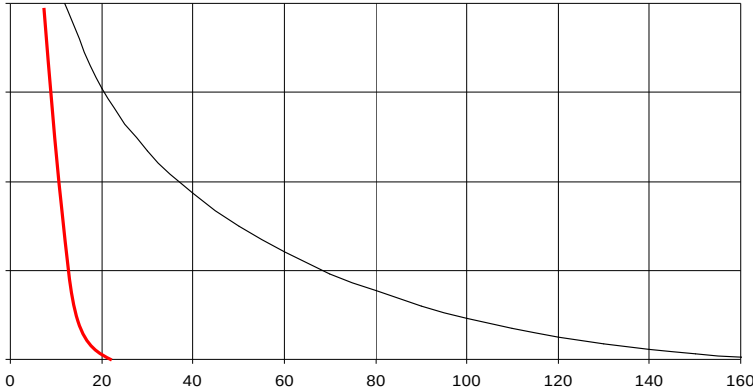


8	(a)(i)	The alpha particles have only limited range in air which is only a few centimetres (~ 5 cm). This is because alpha particles are highly ionizing and lose their kinetic energies very quickly over a short distance due to the interaction with air particles.	B1
	(a)(ii)	N values have a very wide range (from 0 to 10^8). Thus, it is difficult to plot <u>smaller</u> N values if vertical axis is non-logarithmic. OR It compresses the scale so that widely differing values can be shown easily on one graph.	B1
	(b)	 - steep graph at small angles- no scattering at large angles.	A1
	(c)(i)	From Fig. 8.3, when $\theta = 72$, $N = 84000$. $\lg \sin((72/2) / ^\circ) = -0.231$, $\lg 84000 = 4.92$	

1. From Fig. 8.3, when $\theta = 72$, $N = 84000$. (c)(i)2. $\lg \sin((72/2) / ^\circ) = -0.231$, $\lg 84000 = 4.92$		A1 - accurate plotted point A1 - best fit line passes through all plotted points
(c)(ii) Gradient $= (5.28 - 4.7)/(-0.32 - (-0.175))$ $= -4$		M1 A1
(c)(iii) Since $N \propto (\sin(\theta/2))^n$, $\lg N = \lg A + n \lg(\sin(\theta/2))$ where A is a constant. Thus if a graph of $\lg N$ is plotted against $\lg(\sin(\theta/2))$ and a straight line graph is obtained, then n represents the gradient. $n = -4$		M1 A1 - Answer must be an integer
(d) For Au, $92500/79^2 = 14.8$ For Pt, $90000/78^2 = 14.8$ For Ir, $88000/77^2 = 14.8$ For W, $81000/74^2 = 14.8$ Since the constant of proportionality is (approximately) the same to 3 significant figures, N is proportional to Z^2. (based on all the values of constant)		A1 - Correct N values for the 4 data points A1 - showed all the values of the constant A1 - conclusion

(e)	The actual relationship between N and θ is $N = \left(\frac{A'' N_i n L C^2 e^4}{4 r^2 K E^2} \right) Z^2 (\sin(\theta / 2))^{-4}$ where $k = \frac{A'' N_i n L C^2 e^4}{4 r^2 K E^2}$ and A'' = cross sectional area of detector that the scattered alpha particles are incident on N_i = number of incident alpha particles n = number of gold atoms per unit volume in thin gold foil L = thickness of gold foil e = electron charge r = gold foil-to-detector distance KE = kinetic energy of alpha particles $C = \frac{1}{4\pi\epsilon_0}$ = Coulomb's constant, ϵ_0 = permittivity of free space	A1 - any one physical quantities boxed A2 - any two physical quantities boxed
(f)	There would be no (significant) change because the scattering angle θ is (approximately) independent on the mass of the isotope of gold that made up the foil.	B1 B1